Statistics/Data analysis

Title

rdrobust —— Local Polynomial Regression Discontinuity Estimation with Robust Bias-Corrected Confidence Intervals and Inference Procedures.

Syntax

rdrobust *depvar runvar* [*if*] [*in*] [, **c(***#***)** **fuzzy(***fuzzyvar [sharpbw]***)** **deriv(***#***)** **scalepar(***#***)** **p(***#***)** **q(***#***)** **h(***# #***)** **b(***# #***)** **rho(***#***)** **covs(***covars***)** **covs_drop(***covsdropoption***)** **kernel(***kernelfn***)** **weights(***weightsvar***)** **bwselect(***bwmethod***)** **scaleregul(***#***)** **masspoints(***masspointsoption***)** **bwcheck(***#***)** **bwrestrict(***bwropt***)** **stdvars(***stdopt***)** **precision(***precopt***)** **vce(***vcetype [vceopt1 vceopt2]***)** **level(***#***)** **detail** **all**]

Description

rdrobust implements local polynomial Regression Discontinuity (RD) point estimators with robust bias-corrected confidence intervals and inference procedures developed in Calonico, Cattaneo and Titiunik (2014a), Calonico, Cattaneo and Farrell (2018), Calonico, Cattaneo, Farrell and Titiunik (2019), and Calonico, Cattaneo and Farrell (2020). It also computes alternative estimation and inference procedures available in the literature.

Companion commands are: rdbwselect for data-driven bandwidth selection, and rdplot for data-driven RD plots (see Calonico, Cattaneo and Titiunik (2015a) for details).

A detailed introduction to this command is given in Calonico, Cattaneo and Titiunik (2014b), and Calonico, Cattaneo, Farrell and Titiunik (2017). A companion R package is also described in Calonico, Cattaneo and Titiunik (2015b).

Related Stata and R packages useful for inference in RD designs are described in the following website:

https://rdpackages.github.io/

Requires Stata 16 or later.

Options

Estimand

c(*#***)** specifies the RD cutoff for *indepvar*. Default is **c(0)**.

fuzzy(*fuzzyvar [sharpbw]***)** specifies the treatment status variable used to implement fuzzy RD estimation (or Fuzzy Kink RD if **deriv(1)** is also specified). Default is Sharp RD design and hence this option is not used. If the option *sharpbw* is set, the fuzzy RD estimation is performed using a bandwidth selection procedure for the sharp RD model. This option is automatically selected if there is perfect compliance at either side of the threshold.

deriv(*#***)** specifies the order of the derivative of the regression functions to be estimated. Default is **deriv(0)** (for Sharp RD, or for Fuzzy RD if **fuzzy(.)** is also specified). Setting **deriv(1)** results in estimation of a Kink RD design (up to scale), or Fuzzy Kink RD if **fuzzy(.)** is also specified.

scalepar(*#***)** specifies scaling factor for RD parameter of interest. This option is useful when the estimator of interest requires a known multiplicative factor rescaling (e.g., Sharp Kink RD). Default is **scalepar(1)** (no rescaling).

Local Polynomial Regression

p(*#***)** specifies the order of the local polynomial used to construct the point estimator. Default is **p(1)** (local linear regression).

q(#**)** specifies the order of the local polynomial used to construct the bias correction. Default is **q(2)** (local quadratic regression).

h(# #**)** specifies the main bandwidth (*h*) used to construct the RD point estimator. If not specified, bandwidth *h* is computed by the companion command rdbwselect. If two bandwidths are specified, the first bandwidth is used for the data below the cutoff and the second bandwidth is used for the data above the cutoff.

b(# #**)** specifies the bias bandwidth (*b*) used to construct the bias-correction estimator. If not specified, bandwidth *b* is computed by the companion command rdbwselect. If two bandwidths are specified, the first bandwidth is used for the data below the cutoff and the second bandwidth is used for the data above the cutoff.

rho(#**)** specifies the value of *rho*, so that the bias bandwidth *b* equals *b*=*h*/*rho*. Default is **rho(1)** if *h* is specified but *b* is not.

covs(*covars***)** specifies additional covariates to be used for estimation and inference.

covs_drop(*covsdropoption***)** assess collinearity in additional covariates used for estimation and inference. Options **pinv** (default choice) and **invsym** drops collinear additional covariates, differing only in the type of inverse function used. Option **off** omits the check for collinear additional covariates.

kernel(*kernelfn***)** specifies the kernel function used to construct the local-polynomial estimator(s). Options are: **triangular**, **epanechnikov**, and **uniform**. Default is **kernel(triangular)**.

weights(*weightsvar***)** is the variable used for optional weighting of the estimation procedure. The unit-specific weights multiply the kernel function.

Bandwidth Selection

bwselect(*bwmethod***)** specifies the bandwidth selection procedure to be used. By default it computes both *h* and *b*, unless *rho* is specified, in which case it only computes *h* and sets *b*=*h*/*rho*. Options are:
- **mserd** one common MSE-optimal bandwidth selector for the RD treatment effect estimator.
- **msetwo** two different MSE-optimal bandwidth selectors (below and above the cutoff) for the RD treatment effect estimator.
- **msesum** one common MSE-optimal bandwidth selector for the sum of regression estimates (as opposed to difference thereof).
- **msecomb1** for min(**mserd,msesum**).
- **msecomb2** for median(**msetwo,mserd,msesum**), for each side of the cutoff separately.
- **cerrd** one common CER-optimal bandwidth selector for the RD treatment effect estimator.
- **certwo** two different CER-optimal bandwidth selectors (below and above the cutoff) for the RD treatment effect estimator.
- **cersum** one common CER-optimal bandwidth selector for the sum of regression estimates (as opposed to difference thereof).
- **cercomb1** for min(**cerrd,cersum**).
- **cercomb2** for median(**certwo,cerrd,cersum**), for each side of the cutoff separately.

Note: MSE = Mean Square Error; CER = Coverage Error Rate.
Default is **bwselect(mserd)**. For details on implementation see Calonico, Cattaneo and Titiunik (2014a), Calonico, Cattaneo and Farrell (2017), Calonico, Cattaneo and Farrell (2020), and Calonico, Cattaneo, Farrell and Titiunik (2019), and the companion software articles.

scaleregul(#**)** specifies scaling factor for the regularization term added to the denominator of the bandwidth selectors. Setting **scaleregul(0)** removes the regularization term from the bandwidth selectors. Default is **scaleregul(1)**.

masspoints(*masspointsoption***)** checks and controls for repeated observations in the running variable. Options are:
- **off** ignores the presence of mass points.
- **check** looks for and reports the number of unique observations at each side of the cutoff.
- **adjust** controls that the preliminary bandwidths used in the calculations contain a minimal number of unique observations. By default it uses 10 observations, but it can be manually adjusted with the option **bwcheck.**

Default option is **masspoints(adjust).**

bwcheck(*bwcheck***)** if a positive integer is provided, the preliminary bandwidth used in the calculations is enlarged so that at least *bwcheck* unique observations are used.

bwrestrict(*bwropt***)** if set **on**, computed bandwidths are restricted to lie within the range of *runvar*. Default is **on.**

stdvars(*stdopt***)** if set **on**, *depvar* and *runvar* are standardized before computing the bandwidths. Standardization avoids numerical instability in bandwidth selection when *runvar* has very large or very small magnitude. Default is **on.**

precision(*precopt***)** controls the storage precision of internal temporary variables (cluster-id, mass-point counts). Options are **double** (default) and **single;** **single** maps to Stata's **float** storage type. Default is **precision(double).**

Variance-Covariance Estimation

vce(*vcetype [vceopt1 vceopt2]***)** specifies the procedure used to compute the variance-covariance matrix estimator. Options are:
- **vce(nn** *[nnmatch]***)** for heteroskedasticity-robust nearest neighbor variance estimator with *nnmatch* indicating the minimum number of neighbors to be used.
- **vce(hc0)** for heteroskedasticity-robust plug-in residuals variance estimator without weights.
- **vce(hc1)** for heteroskedasticity-robust plug-in residuals variance estimator with *hc1* weights.
- **vce(hc2)** for heteroskedasticity-robust plug-in residuals variance estimator with *hc2* weights.
- **vce(hc3)** for heteroskedasticity-robust plug-in residuals variance estimator with *hc3* weights.
- **vce(cluster** *clustervar***)** for cluster-robust plug-in residuals variance estimation with degrees-of-freedom weights; equivalent to **vce(cr1** *clustervar***).**
- **vce(cr1** *clustervar***)** for cluster-robust plug-in residuals variance estimator with degrees-of-freedom correction ((n-1)/(n-k)*g/(g-1)).
- **vce(cr2** *clustervar***)** for the Bell-McCaffrey (2002) bias-reduced cluster-robust variance estimator (CRV2).
- **vce(cr3** *clustervar***)** for the Pustejovsky-Tipton (2018) cluster-robust variance estimator (CRV3), approximately unbiased with few clusters.

The CR2/CR3 leverage correction applies to both the conventional and the robust bias-corrected standard errors, including when the point-estimation bandwidth h differs from the bias-correction bandwidth b; in that case the cluster leverage is computed from the bias (b) regression.

Default is **vce(nn 3).**

level(*#***)** specifies confidence level for confidence intervals. Default is **level(95).**

Other Options

all if specified, **rdrobust** reports three different procedures:
(i) conventional RD estimates with conventional variance estimator.
(ii) bias-corrected RD estimates with conventional variance estimator.
(iii) bias-corrected RD estimates with robust variance estimator.

detail if specified, **rdrobust** reports the original output up to version 9.2.0

Example: Cattaneo, Frandsen and Titiunik (2015) Incumbency Data

Setup
    . use rdrobust_senate.dta

Robust RD Estimation using MSE bandwidth selection procedure
    . rdrobust vote margin

Robust RD Estimation with both bandwidths set to 15
    . rdrobust vote margin, h(15)

Other generic examples (**y** outcome variable, **x** running variable, **t** treatment take-up indicator):

Estimation for Sharp RD designs
    . rdrobust y x, deriv(0)

Estimation for Sharp Kink RD designs
    . rdrobust y x, deriv(1)

Estimation for Fuzzy RD designs
    . rdrobust y x, fuzzy(t)

Estimation for Fuzzy Kink RD designs
    . rdrobust y x, fuzzy(t) deriv(1)

Stored results

rdrobust stores the following in **e()**:

Scalars

e(N)	original number of observations
e(N_l)	original number of observations to the left of the cutoff
e(N_r)	original number of observations to the right of the cutoff
e(N_h_l)	effective number of observations (given by the bandwidth h_l) used to the left of the cutoff
e(N_h_r)	effective number of observations (given by the bandwidth h_r) used to the right of the cutoff
e(N_b_l)	effective number of observations (given by the bandwidth b_l) used to the left of the cutoff
e(N_b_r)	effective number of observations (given by the bandwidth b_r) used to the right of the cutoff
e(c)	cutoff value
e(p)	order of the polynomial used for estimation of the regression function
e(q)	order of the polynomial used for estimation of the bias of the regression function estimator
e(h_l)	bandwidth used for estimation of the regression function below the cutoff
e(h_r)	bandwidth used for estimation of the regression function above the cutoff
e(b_l)	bandwidth used for estimation of the bias of the regression function estimator below the cutoff
e(b_r)	bandwidth used for estimation of the bias of the regression function estimator above the cutoff
e(tau_cl)	conventional local-polynomial RD estimate
e(tau_cl_l)	conventional local-polynomial left estimate
e(tau_cl_r)	conventional local-polynomial right estimate
e(tau_bc)	bias-corrected local-polynomial RD estimate
e(tau_bc_l)	bias-corrected local-polynomial left estimate
e(tau_bc_r)	bias-corrected local-polynomial right estimate
e(se_tau_cl)	conventional standard error of the local-polynomial RD estimator
e(se_tau_rb)	robust standard error of the local-polynomial RD estimator
e(tau_T_cl)	conventional first-stage estimate (fuzzy RD only)
e(tau_T_cl_l)	conventional first-stage left estimate (fuzzy RD only)
e(tau_T_cl_r)	conventional first-stage right estimate (fuzzy RD only)
e(tau_T_bc)	bias-corrected first-stage estimate (fuzzy RD only)
e(tau_T_bc_l)	bias-corrected first-stage left estimate (fuzzy RD only)
e(tau_T_bc_r)	bias-corrected first-stage right estimate (fuzzy RD only)

e(se_tau_T_cl) conventional standard error of the first-stage estimator (fuzzy RD only)
e(se_tau_T_rb) robust standard error of the first-stage estimator (fuzzy RD only)
e(bias_l) estimated bias for the local-polynomial RD estimator below the cutoff
e(bias_r) estimated bias for the local-polynomial RD estimator above the cutoff
e(level) confidence level
e(ci_l_cl) lower endpoint of the conventional CI
e(ci_r_cl) upper endpoint of the conventional CI
e(ci_l_rb) lower endpoint of the robust bias-corrected CI
e(ci_r_rb) upper endpoint of the robust bias-corrected CI
e(pv_cl) p-value of the conventional test
e(pv_bc) p-value of the bias-corrected test (uses conventional SE)
e(pv_rb) p-value of the robust bias-corrected test
e(n_clust) number of clusters (only when **vce(cluster**|**cr1**|**cr2**|**cr3 ...)** is specified)

Macros
e(cmd) **rdrobust**
e(cmdline) command as typed
e(title) title for the output (**Sharp**/**Fuzzy RD estimates**, optionally **(covariate-adjusted)**)
e(depvar) name of dependent (outcome) variable
e(runningvar) name of running variable
e(clustvar) name of cluster variable
e(covs) names of additional covariates
e(vce_select) vcetype specified in vce() (raw option name, e.g. "nn", "hc3", "cr3")
e(vce_type) display label for variance-covariance estimator (e.g. "NN", "HC3", "CR3")
e(bwselect) bandwidth selection choice
e(kernel) kernel choice
e(ci_rb) formatted robust bias-corrected CI string (e.g. **[3.989 ; 11.021]**); see **e(ci_l_rb)** / **e(ci_r_rb)** for numeric endpoints or **e(ci)** for the full matrix
e(precision) storage precision selected via **precision()**: **double** (default) or **single**

Matrices
e(b) 1x3 coefficient vector with column names **Conventional** (tau_cl), **Bias-corrected** (tau_bc), and **Robust** (tau_bc)
e(V) 3x3 block-diagonal variance-covariance matrix for **e(b)**: **V[Conventional,Conventional]** = **V[Bias-corrected,Bias-corrected]** = se_tau_cl^2 and **V[Robust,Robust]** = se_tau_rb^2. The Bias-corrected pairing of tau_bc with the conventional SE is not a recommended inferential object (CCT 2014); use **_b[Robust]** and **_se[Robust]** for the RBC inference.
e(ci) 3x2 confidence-interval matrix; rows **Conventional** / **Bias-corrected** / **Robust**, columns **ll** / **ul**
e(bws) 2x2 bandwidth matrix; rows **h** / **b**, columns **left** / **right**. Mirrors **fit$bws** (R) and **fit.bws** (Python).
e(beta_Y_p_r) conventional p-order local-polynomial estimates to the right of the cutoff for the outcome variable
e(beta_Y_p_l) conventional p-order local-polynomial estimates to the left of the cutoff for the outcome variable
e(beta_T_p_r) conventional p-order local-polynomial estimates to the right of the cutoff for the first stage (fuzzy RD)
e(beta_T_p_l) conventional p-order local-polynomial estimates to the left of the cutoff for the first stage (fuzzy RD)
e(coef_covs) coefficients of the additional covariates, only returned when **covs()** are used
e(V_cl_r) conventional variance-covariance matrix to the right of the cutoff
e(V_cl_l) conventional variance-covariance matrix to the left of the cutoff
e(V_rb_r) robust variance-covariance matrix to the right of the cutoff
e(V_rb_l) robust variance-covariance matrix to the left of the cutoff

References

Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2020. Optimal Bandwidth Choice for Robust Bias Corrected Inference in Regression Discontinuity Designs. *Econometrics Journal* 23(2): 192-210.

Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2018. On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference. *Journal of the American Statistical Association* 113(522): 767-779.

Calonico, S., M. D. Cattaneo, M. H. Farrell, and R. Titiunik. 2019. Regression Discontinuity Designs using Covariates. *Review of Economics and Statistics*, 101(3): 442-451.

Calonico, S., M. D. Cattaneo, M. H. Farrell, and R. Titiunik. 2017. rdrobust: Software for Regression Discontinuity Designs. *Stata Journal* 17(2): 372-404.

Calonico, S., M. D. Cattaneo, and R. Titiunik. 2014a. Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs. *Econometrica* 82(6): 2295-2326.

Calonico, S., M. D. Cattaneo, and R. Titiunik. 2014b. Robust Data-Driven Inference in the Regression-Discontinuity Design. *Stata Journal* 14(4): 909-946.

Calonico, S., M. D. Cattaneo, and R. Titiunik. 2015a. Optimal Data-Driven Regression Discontinuity Plots. *Journal of the American Statistical Association* 110(512): 1753-1769.

Calonico, S., M. D. Cattaneo, and R. Titiunik. 2015b. rdrobust: An R Package for Robust Nonparametric Inference in Regression-Discontinuity Designs. *R Journal* 7(1): 38-51.

Cattaneo, M. D., B. Frandsen, and R. Titiunik. 2015. Randomization Inference in the Regression Discontinuity Design: An Application to Party Advantages in the U.S. Senate. *Journal of Causal Inference* 3(1): 1-24.

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